

L9: Number of “Close Proximity” Contacts

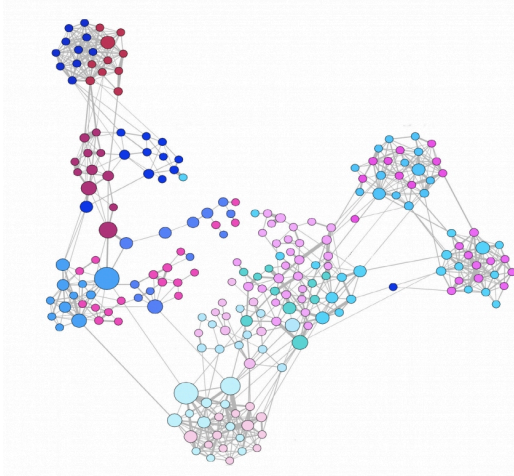


Figure 10.16 from Network Science by Albert-László Barabási .

For airborne pathogens and respiratory diseases such as COVID-19, what matters more is the number of individuals we are in close proximity to. This cannot be measured with surveys but it can be measured with wireless technology such as RFID badges (*Radio Frequency Identification*) . Various studies have provided volunteers with RFID badges and asked them to wear them throughout the whole day (e.g., on university campuses, dorms, gyms) .

The visualization at the left refers to a network of contacts, mapped using RFID technology, between 232 students and 10 teachers across 10 classes in a school. It is also easy to see that there is a very strong community structure in this network, most likely associated with the

different classes the students attend.

A common conclusion from these studies is that the number of people we come close to varies greatly across individuals. Most of us come physically close to only a small number of specific other people but some individuals interact with hundreds of other people in their daily life. RFID technology can also give us information about the duration of these interactions, which is also a very important factor in the transmission of a pathogen from an infected to a susceptible individual.

The statistical distribution of these durations is also heavy-tailed, typically following a power-law, meaning that most of our face-to-face interactions are very brief (e.g., saying *hi* at a corridor) but few interactions last for hours – and typically those are the most dangerous for the transmissions of viruses such as COVID-19, H1N1, influenza, etc.